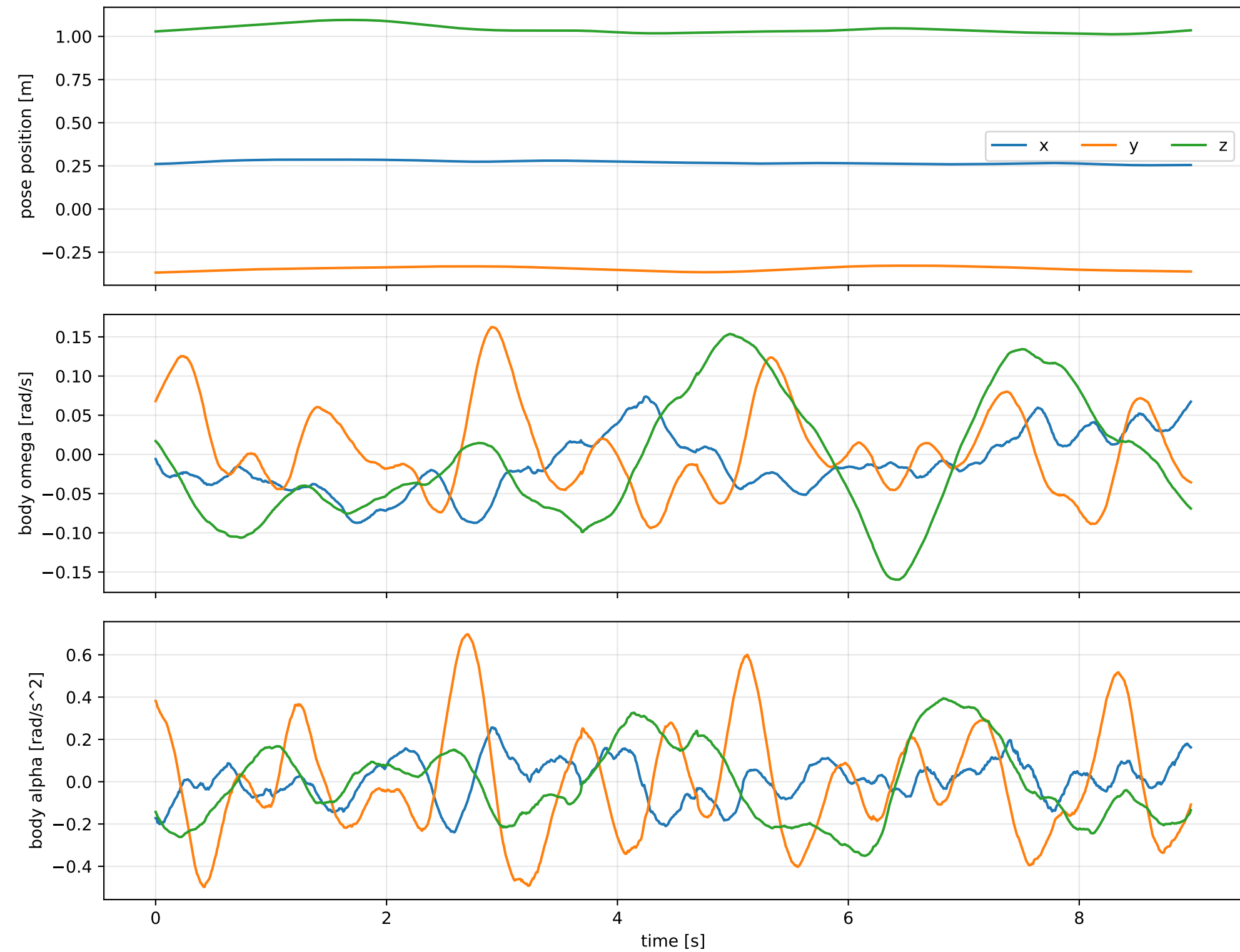
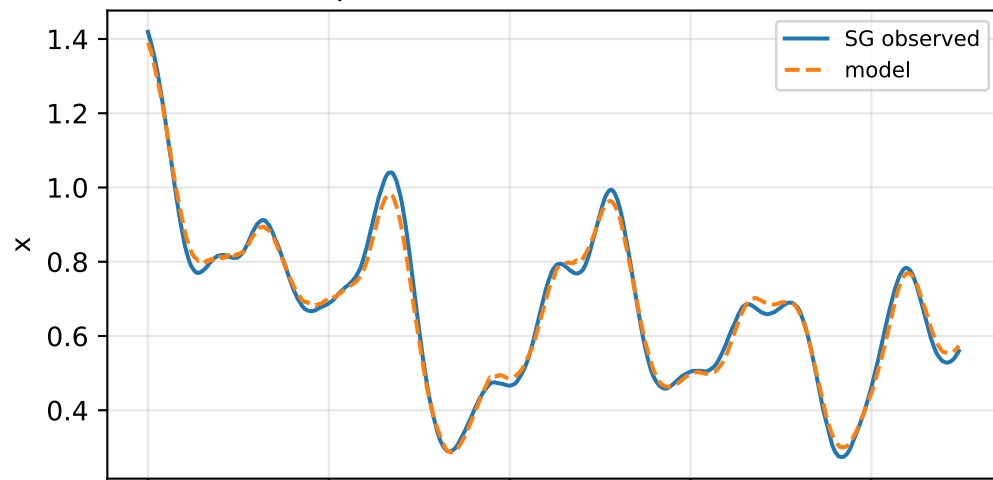


cov_block_s_alpha: SG trajectory and derived motion

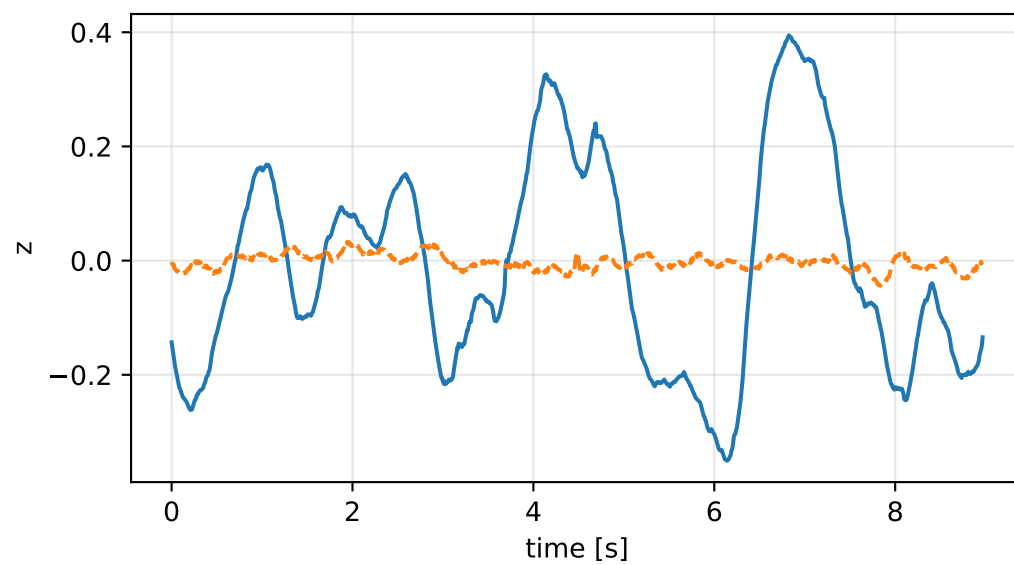
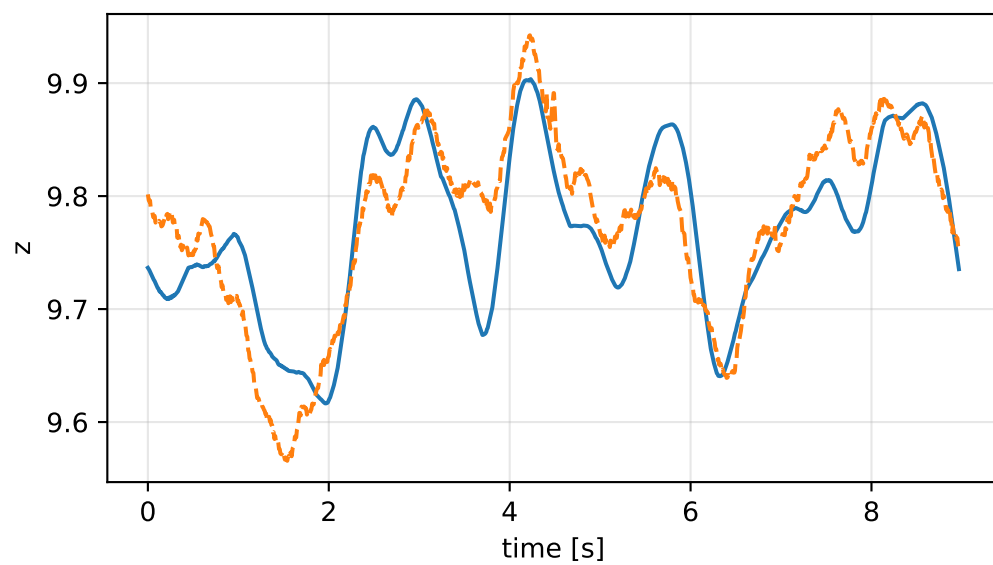
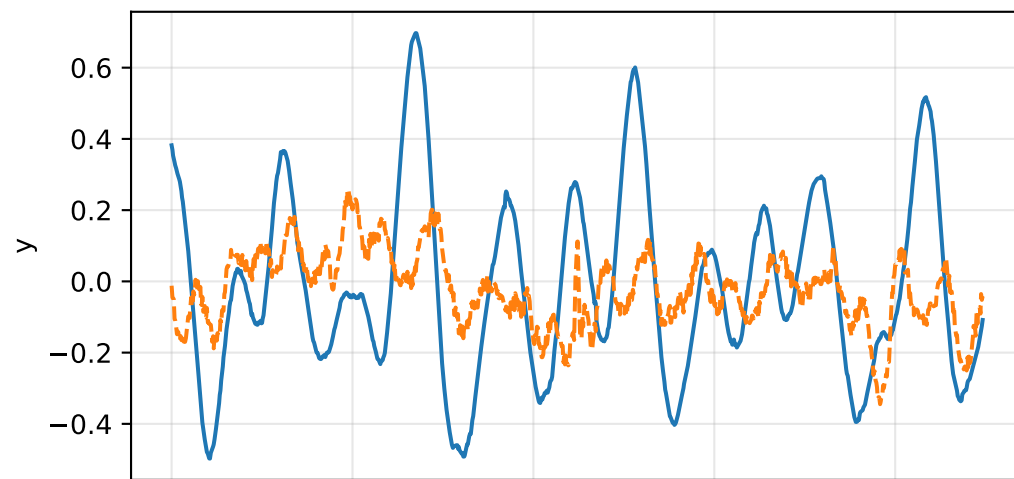
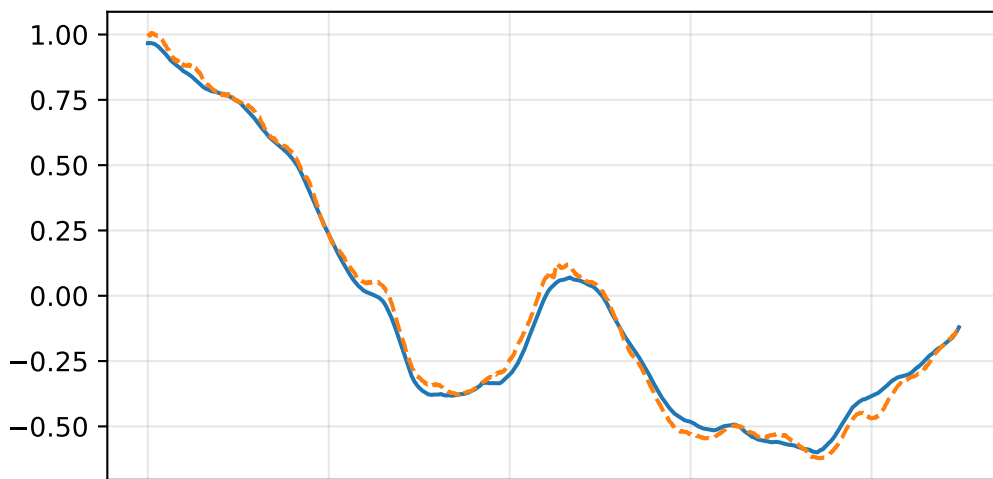
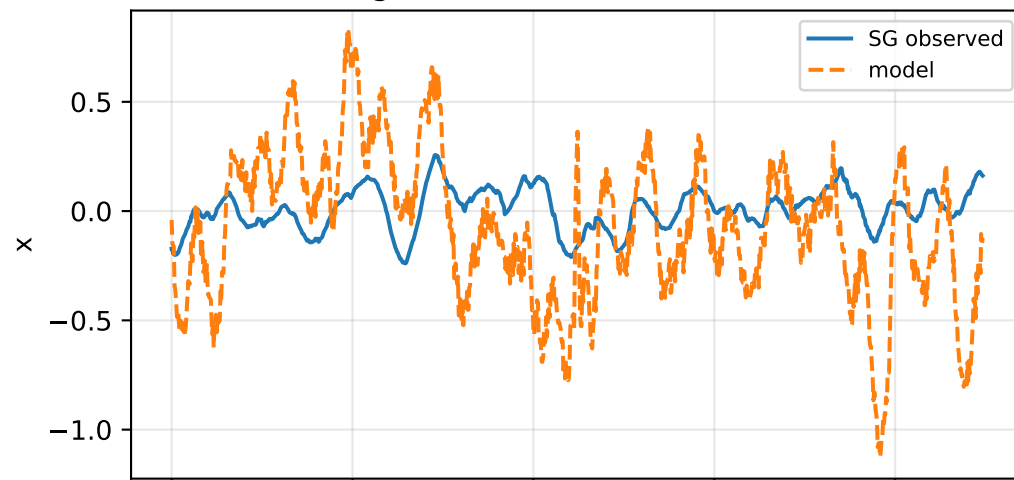


cov_block_s_alpha: acceleration residual objective

specific acceleration [m/s^2]

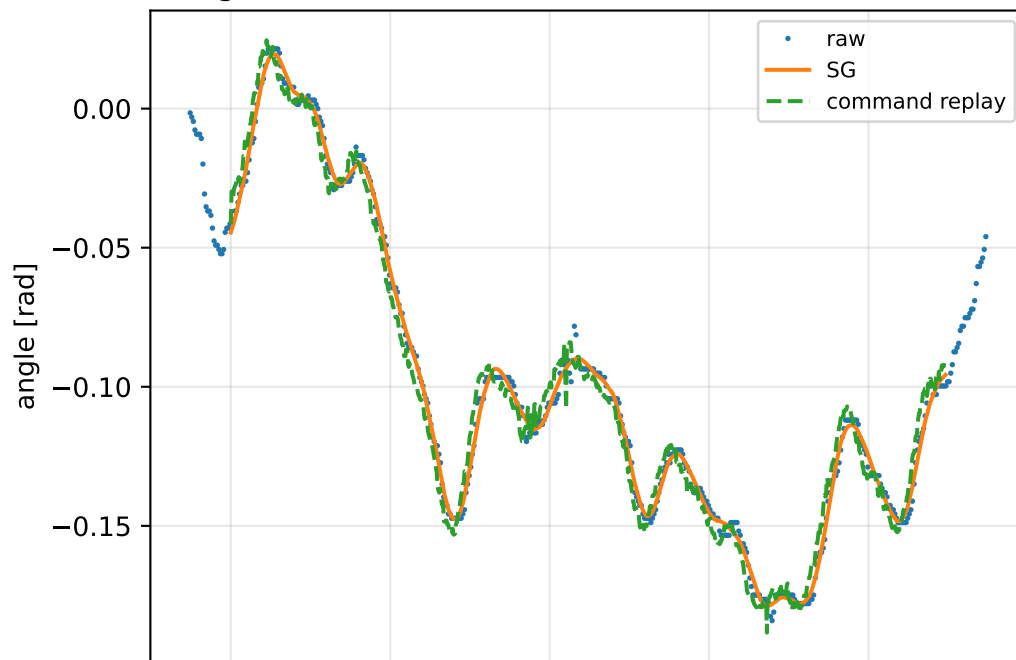


angular acceleration [rad/s^2]

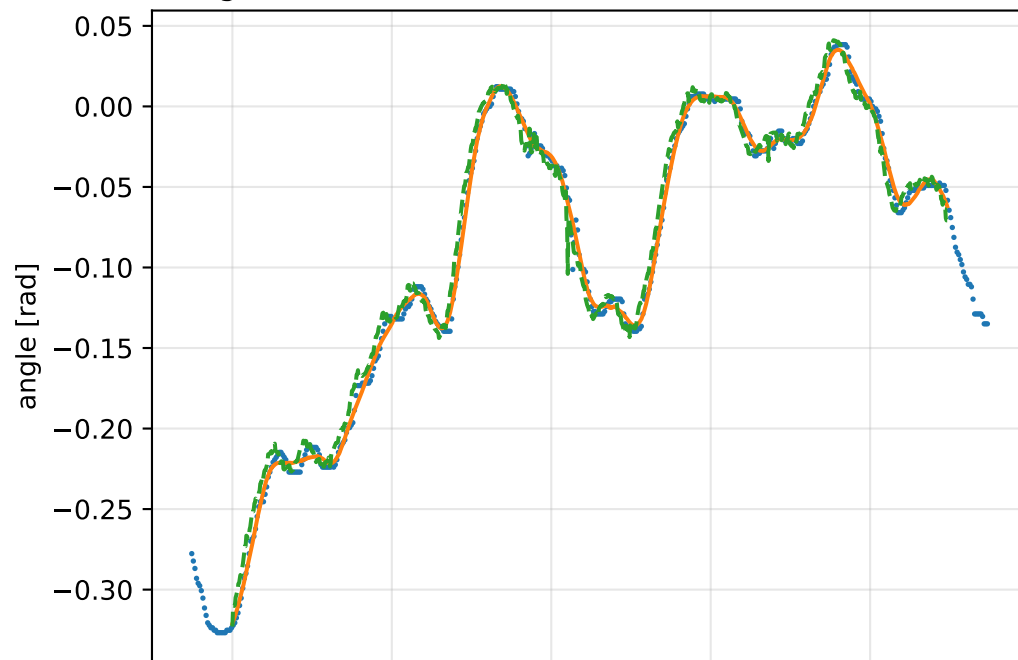


cov_block_s_alpha: gimbal measurement audit (objective=measured_sg)

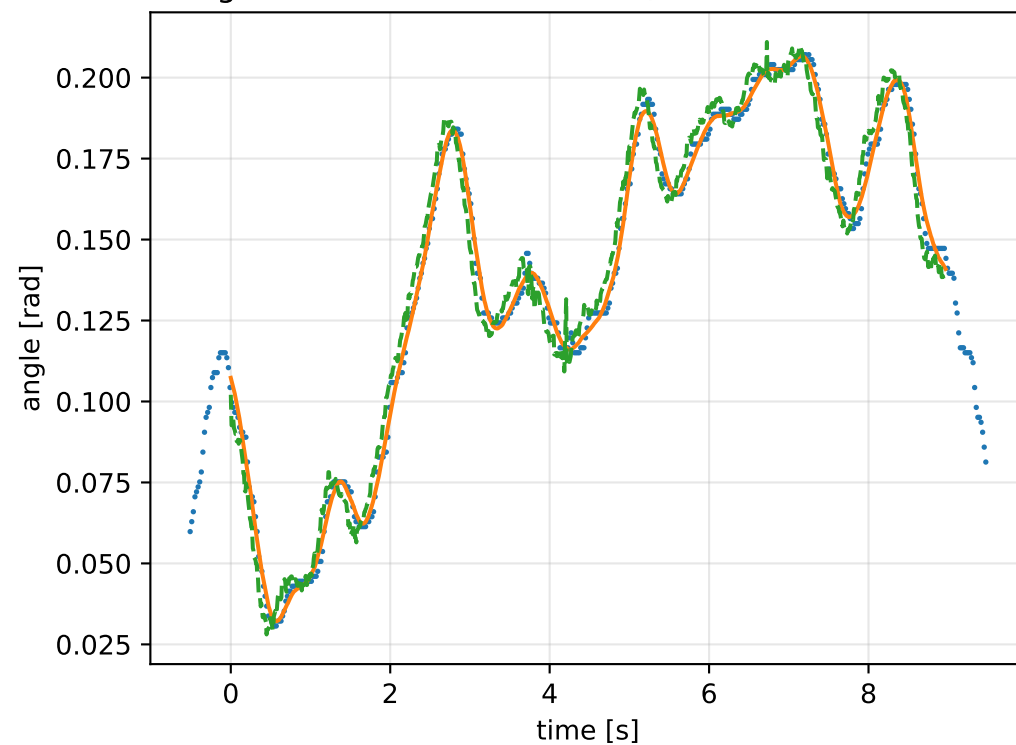
gimbal 1: RMSE=0.006345, max=0.01494 rad



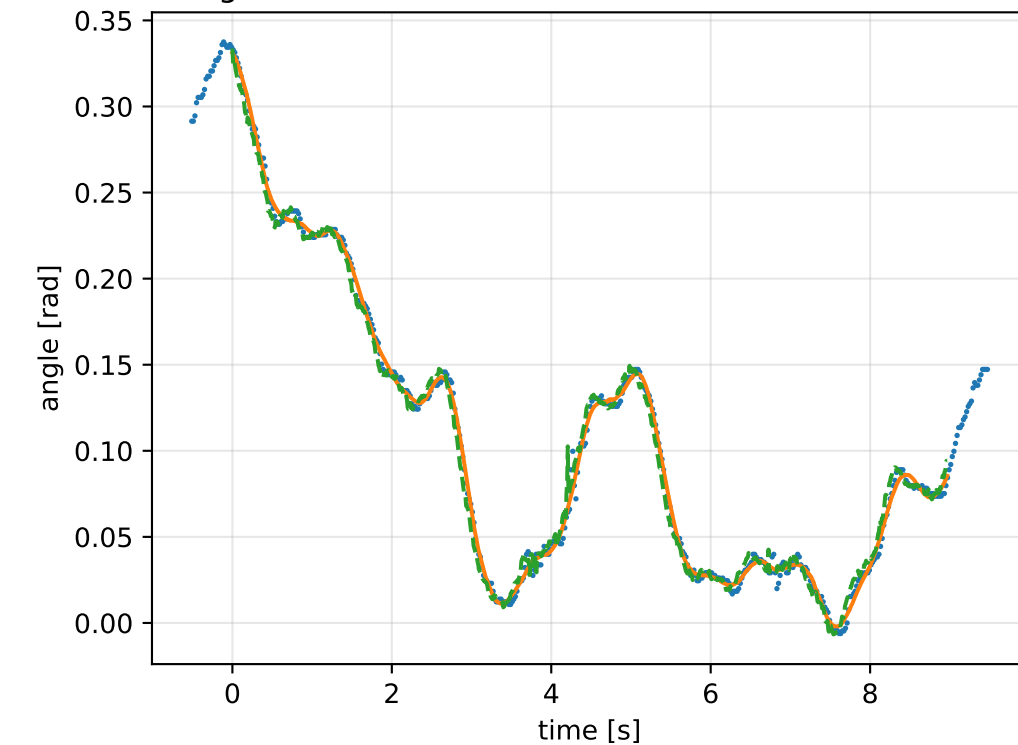
gimbal 2: RMSE=0.008835, max=0.04462 rad



gimbal 3: RMSE=0.007019, max=0.01611 rad

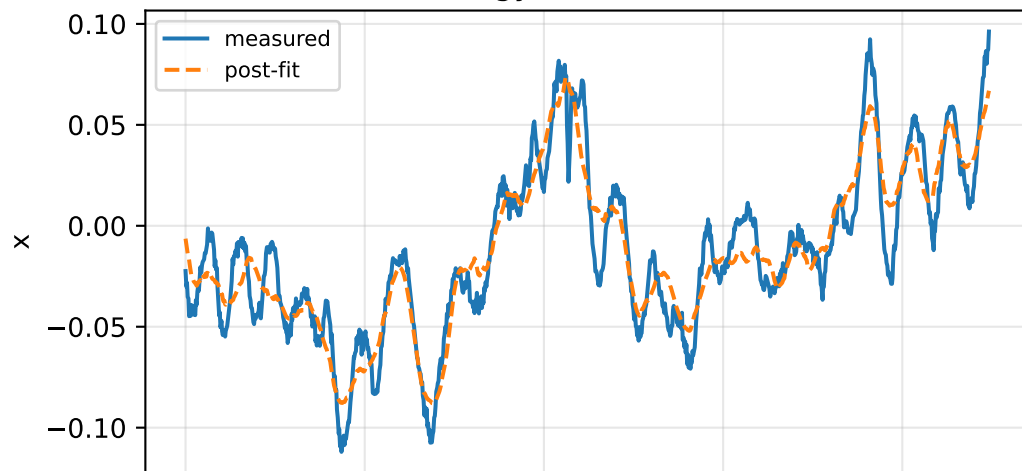


gimbal 4: RMSE=0.006321, max=0.03642 rad

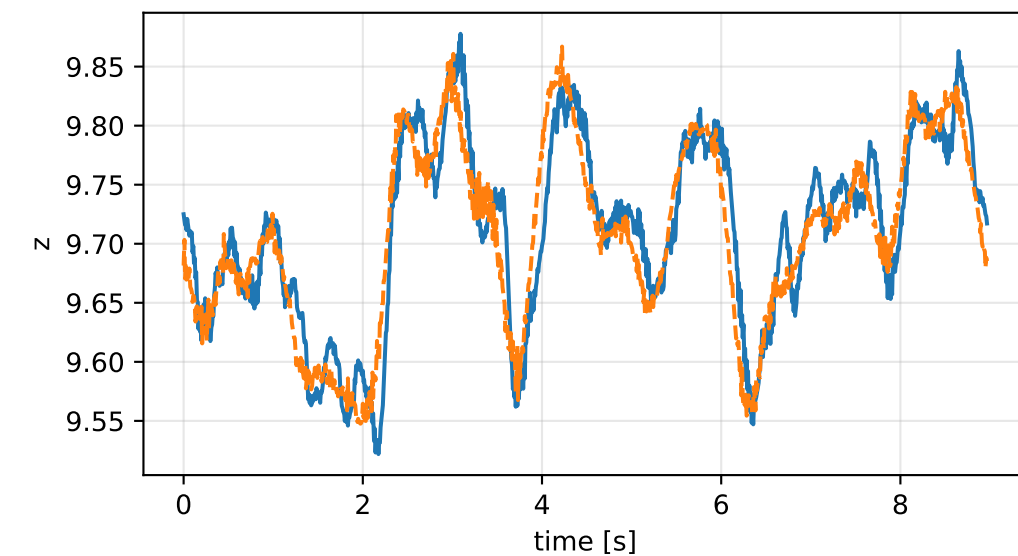
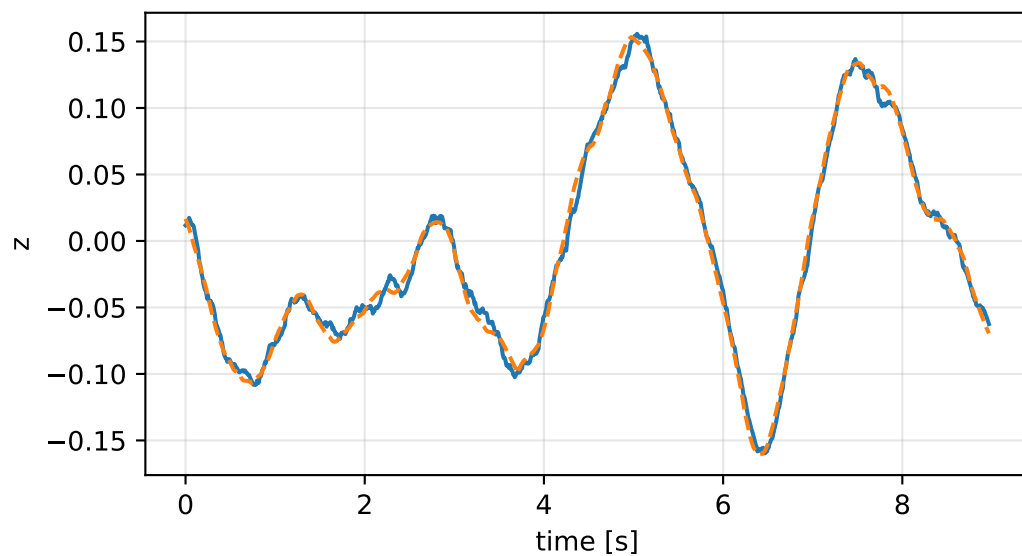
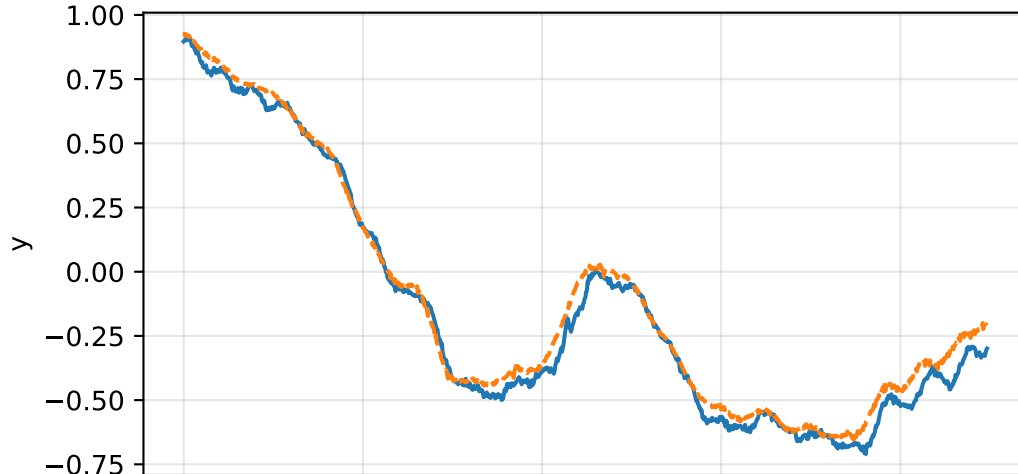
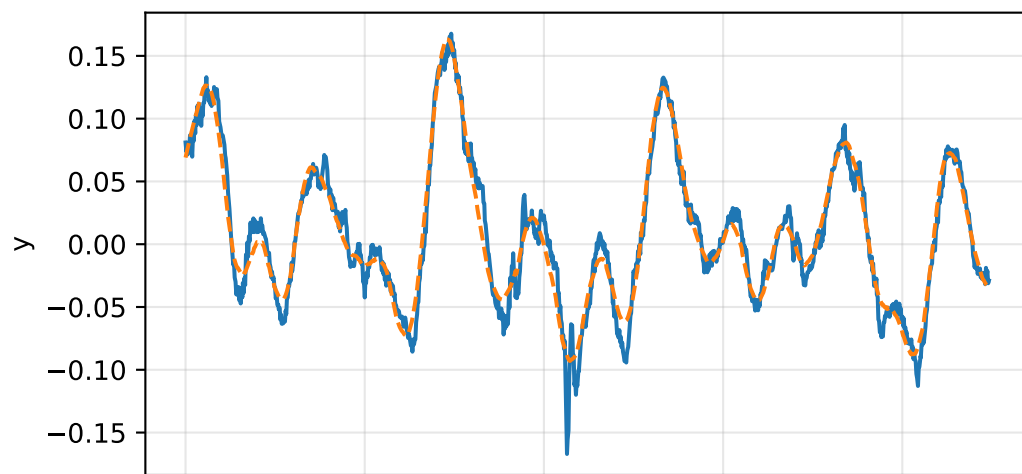
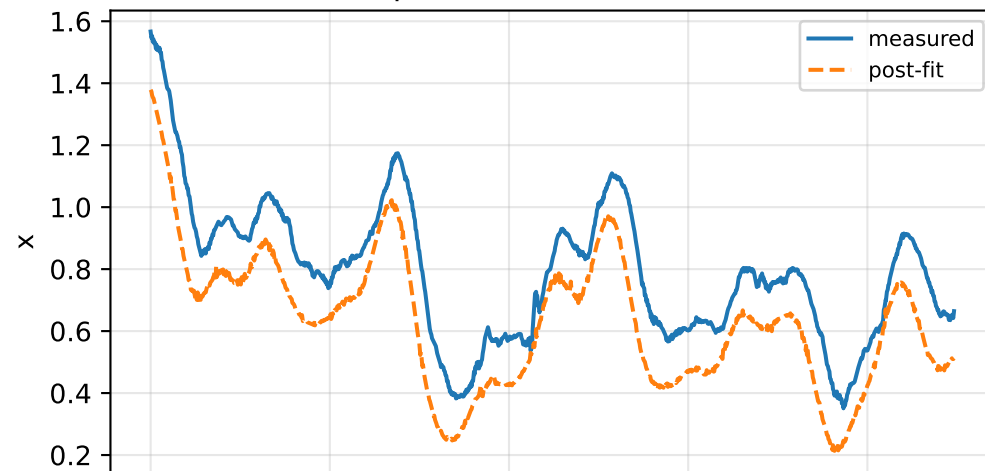


cov_block_s_alpha: IMU comparison (diagnostic only)

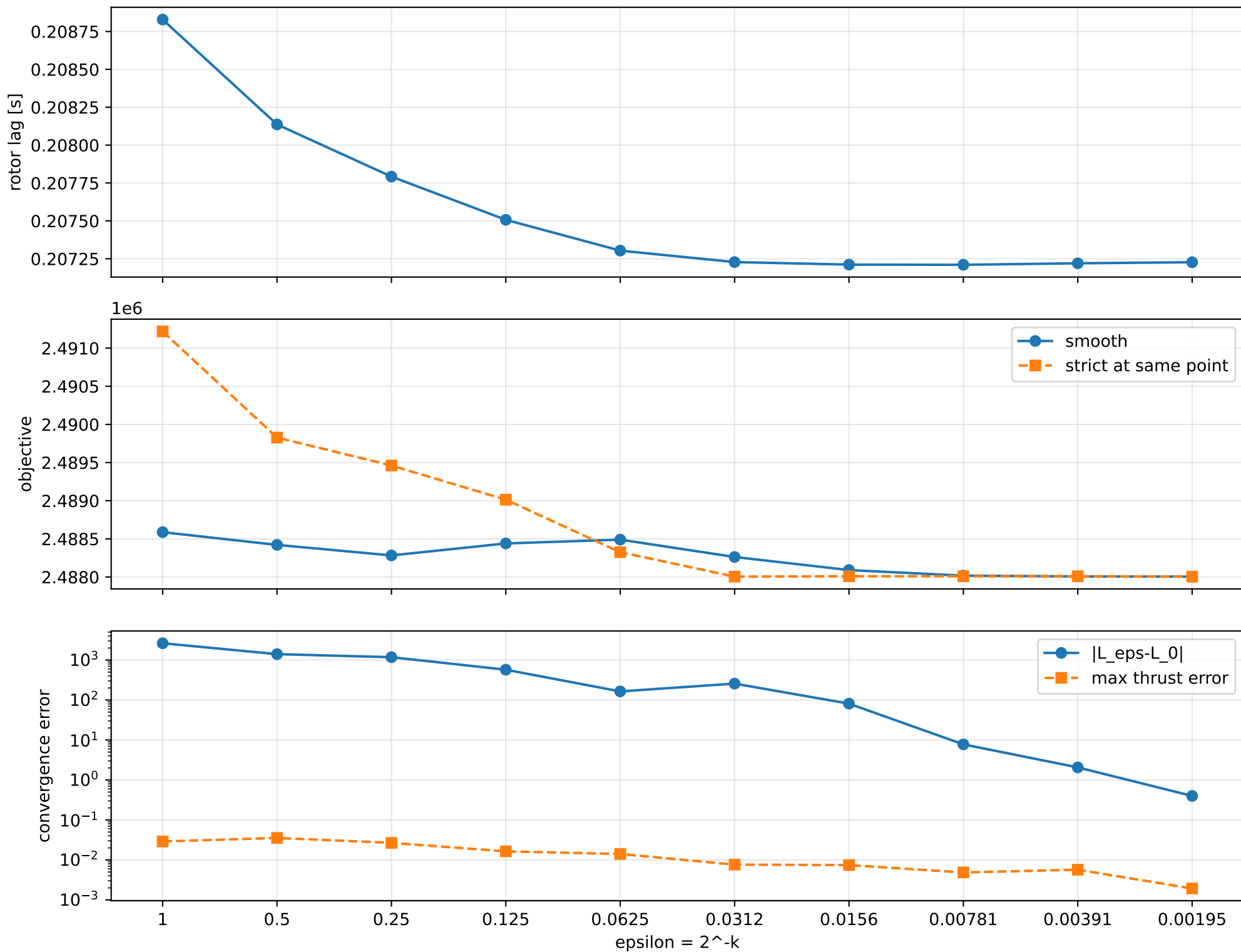
gyro [rad/s]



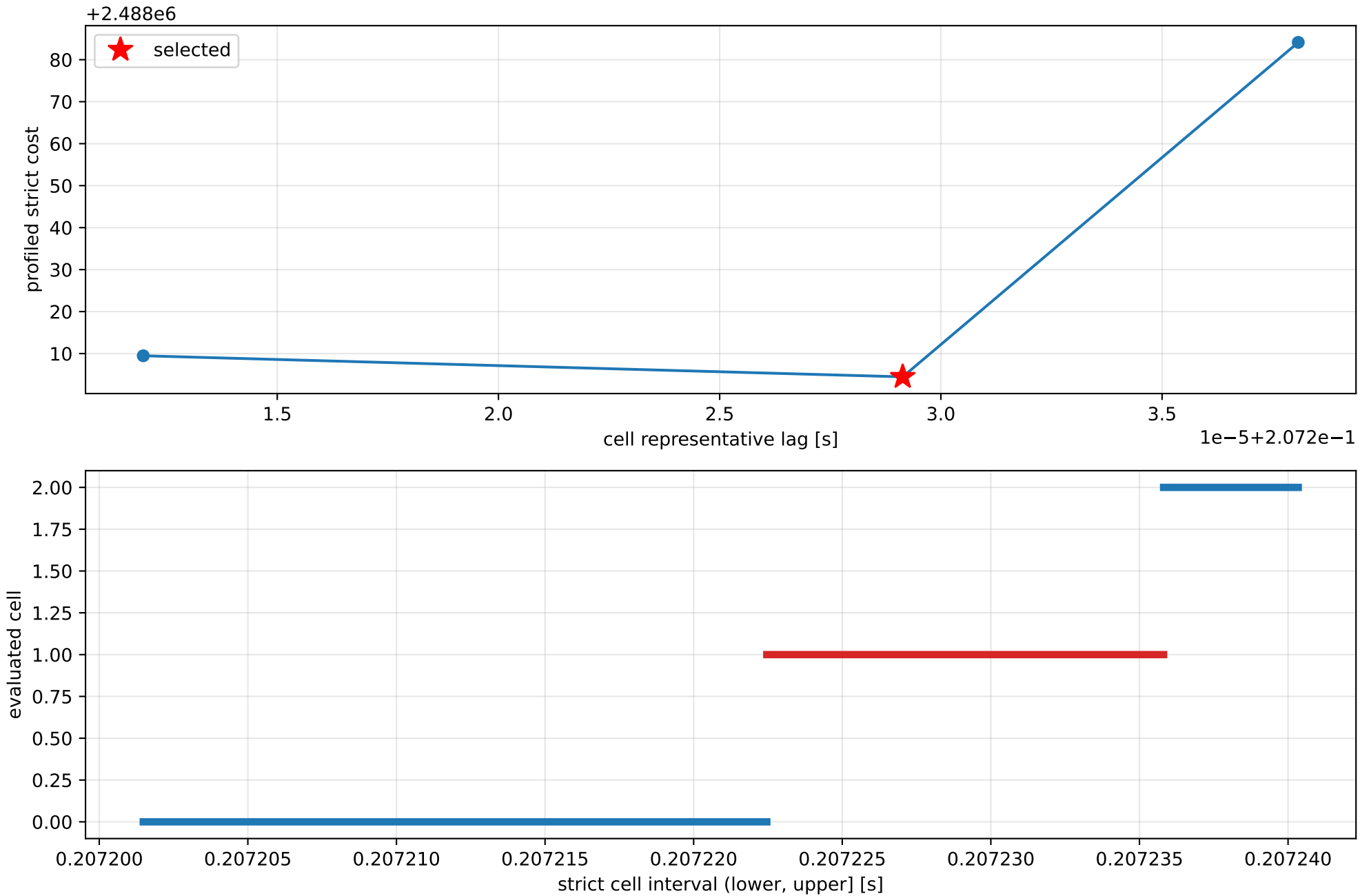
specific force [m/s^2]



cov_block_s_alpha: smooth-to-strict continuation



cov_block_s_alpha: exact strict-ZOH lag cells



selected (0.207222462, 0.207235813] s; final smooth 0.207226907 s; data boundary=False

cov_block_s_alpha: parameter estimate

Case: cov_block_s_alpha

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 0.727050977

inertia [kg m²]:

```
[[ 23650.40413438 -76027.71687075 -9750.9454669 ]
 [-76027.71687075 248805.47194831 -2984.20250764]
 [ -9750.9454669  -2984.20250764 271690.38765628]]
```

CoG body [m]: [-1.09782739 -0.51053384 -0.01788581]

force effectiveness: [0.34791004 0.53866807 0.12721948 0.02336417]

scale-free J/m [m²]:

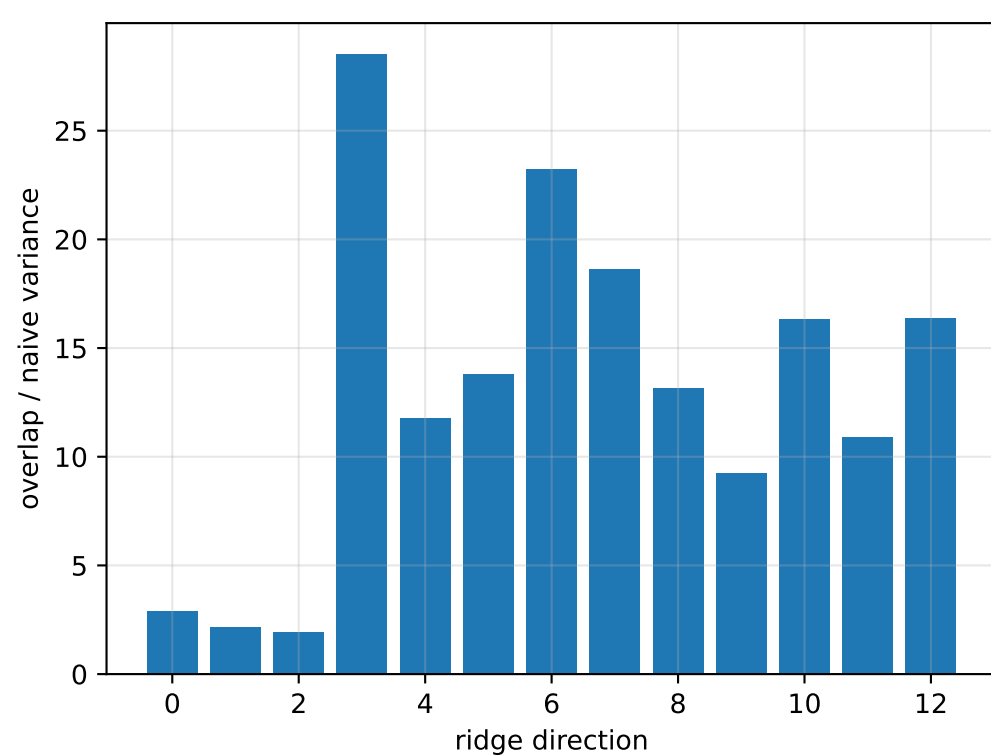
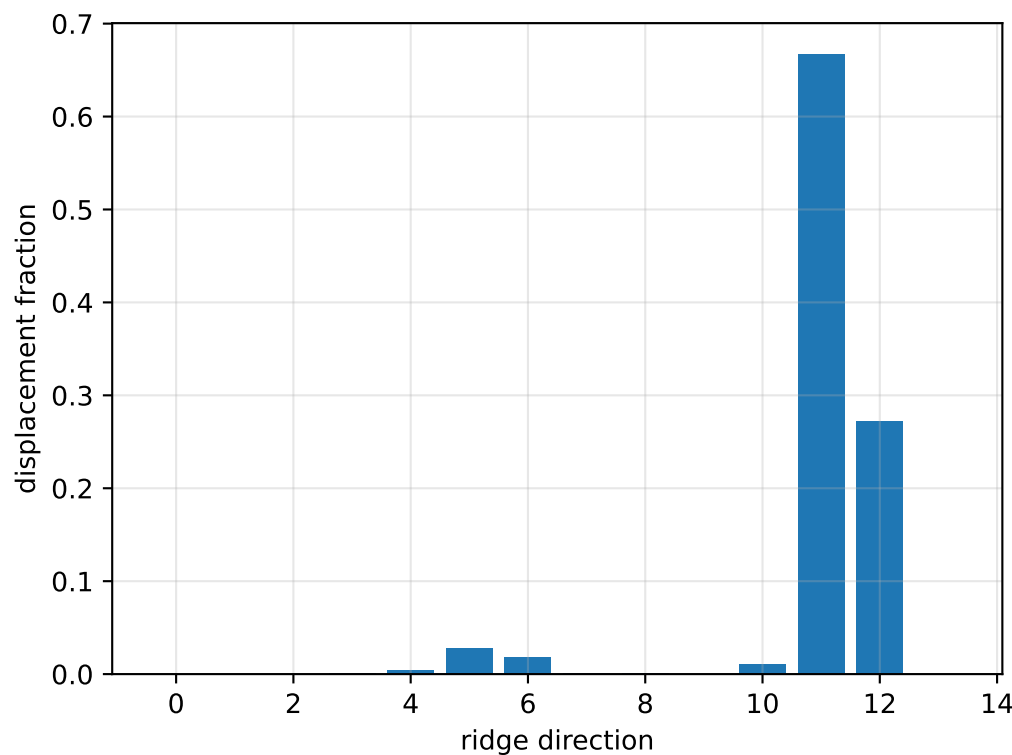
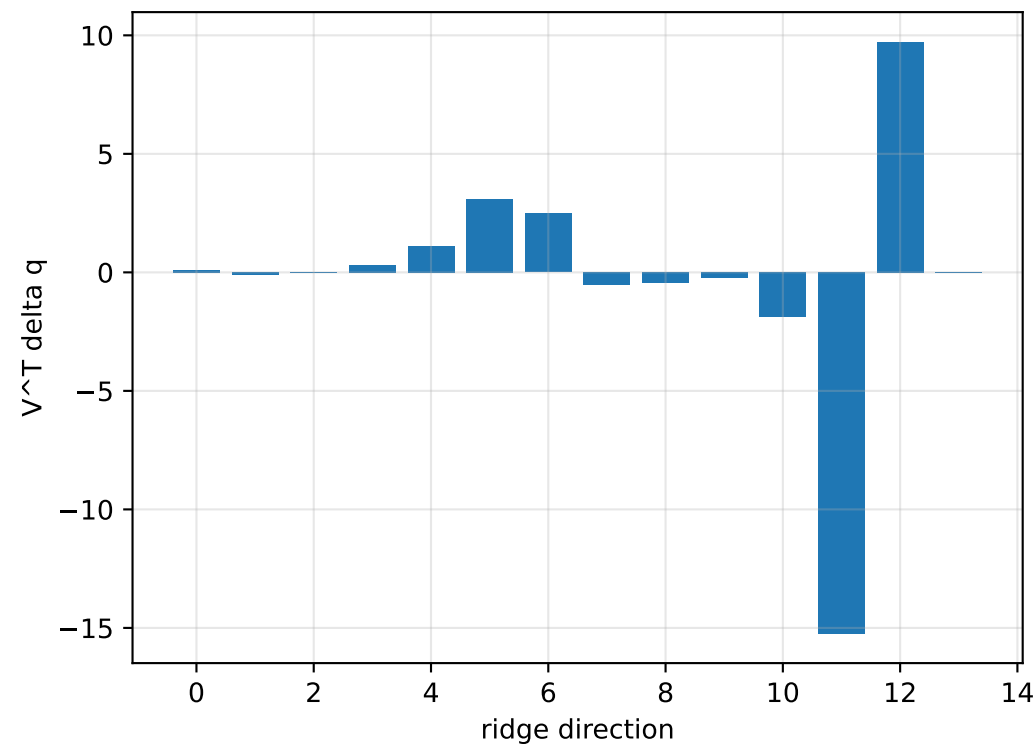
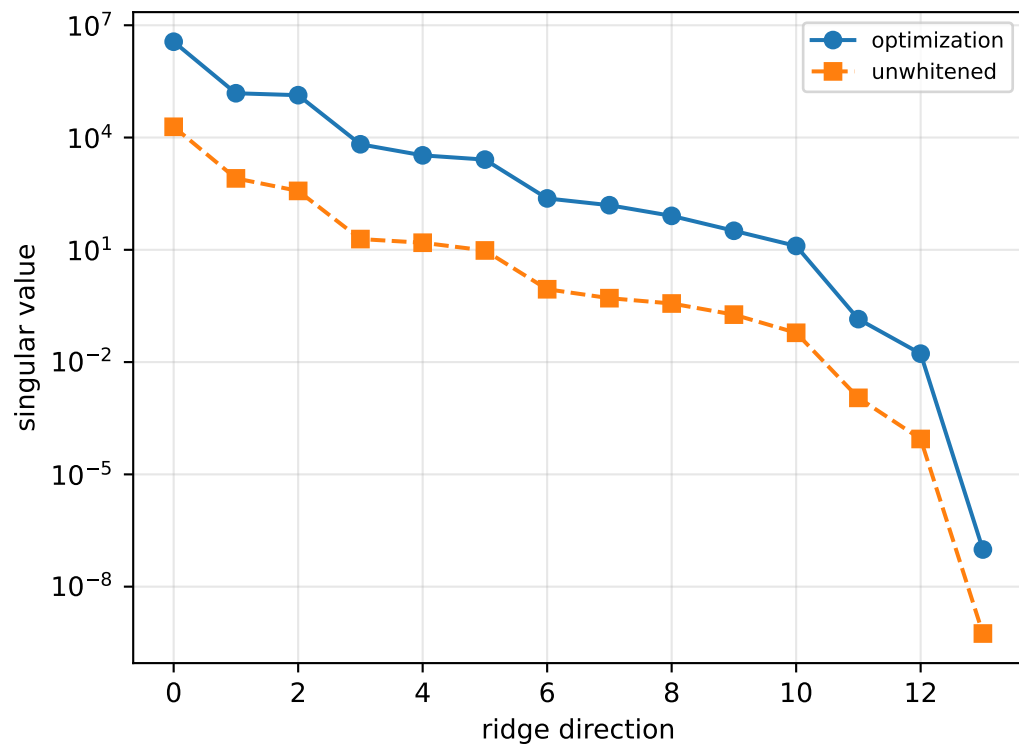
```
[[ 32529.22407625 -104569.99483146 -13411.63932108]
 [-104569.99483146 342211.86675784 -4104.52994833]
 [ -13411.63932108 -4104.52994833 373688.22322095]]
```

scale-free f/m: [0.47852221 0.7408945 0.17498014 0.03213553]

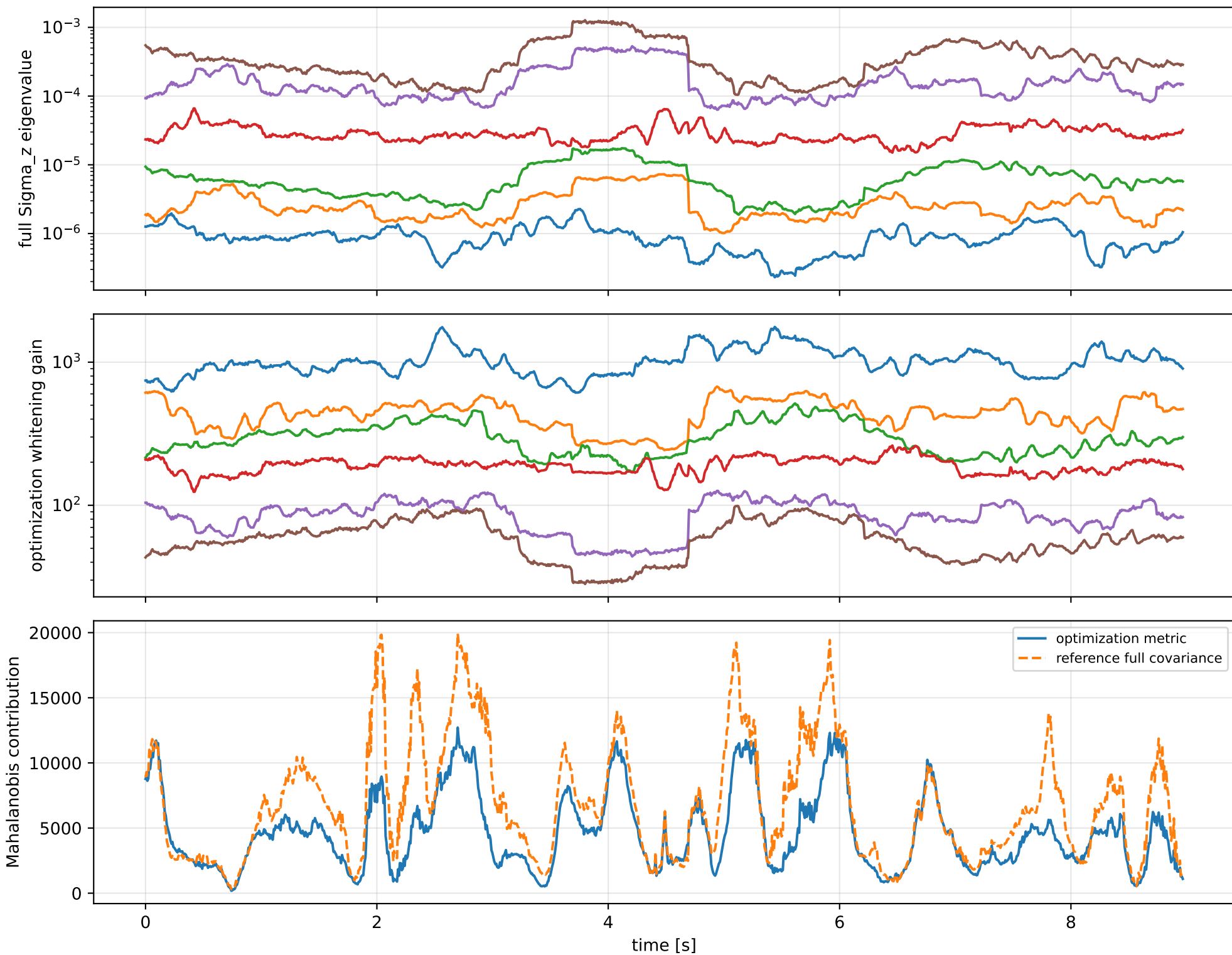
rotor lag cell: (0.207222462, 0.207235813] s

representative: 0.207229137 s

cov_block_s_alpha: ridge spectrum (rank 13, nullity 1, || $\mathbf{v}$ ||=1.62e-06)



cov_block_s_alpha: optimization vs reference covariance



cov_block_s_alpha: wrench / closure (force max 2.21e-15, torque max 5.82e-11)

